

Cost analysis

If you can't measure it, you can't improve it. — Peter Drucker

Before we dive into optimisation, it's essential to analyse the cost drivers. Fortunately, Google Cloud Platform provides a detailed monthly billing report. The 'Cloud Billing' feature offers transparent visibility into costs incurred by various cloud services, including compute, storage, network, and more. The 'Reports' section in 'Cloud Billing' is a good starting point, as it provides various types of reports on the costs incurred by a Google project. It includes 'Group by' and 'Filter' options to view the costs by service, SKU, etc. By looking at this data, we

can quickly identify which service and specific SKU is contributing to the high costs. We can then look at what options are available to optimise costs for that service, picking the top cost contributor to optimise what might give us better ROI in terms of effort versus cost saved.

Optimisation strategies

Once the service(s) contributing to the higher costs are identified, the next step is to look at the options available to optimise costs. The right option will depend on the application and its use case of the service. Also, a typical enterprise deploys multiple environments – development (dev), test, staging, production (prod), etc. Optimisation strategies vary by environment as resource uptime and responsiveness requirements differ across

environments. Let us look at some of the widely used GCP services, the factors that influence associated costs, and the options available to optimise costs in different scenarios.

Compute

Compute costs are dependent on the resources (CPU, memory, storage) allocated and the machine type, which plays a significant role. So, choosing the optimal machine type depending on the environment and the application use case becomes crucial.

• Non-production environments

Machine types: Production environments may need specialised machine types (like compute-optimised) or the latest types (like C4/N4 series) that have the

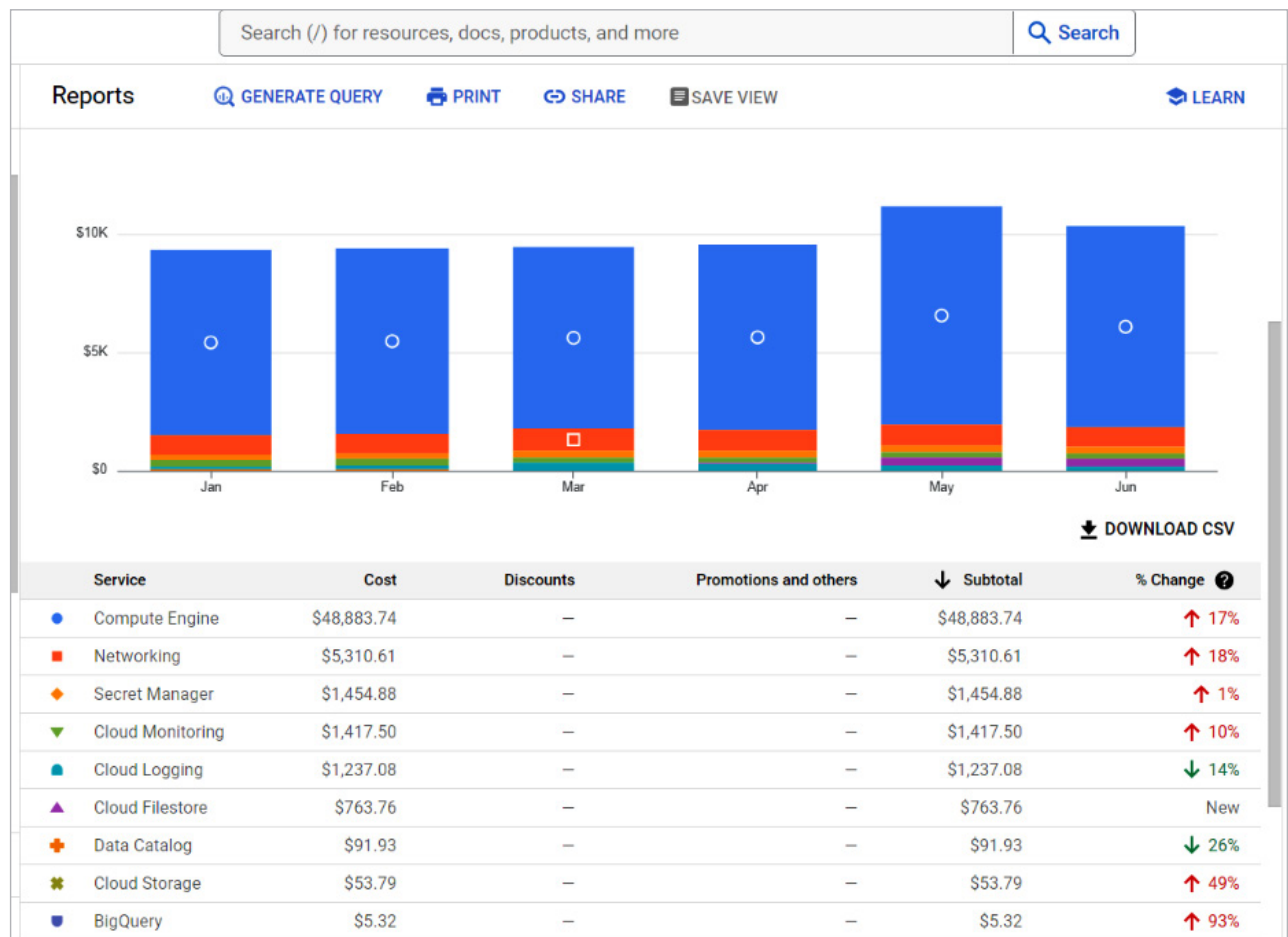


Figure 1: GCP cloud billing reports